CHAPTER 1

This chapter introduces the study by presenting the background and context, defining the research problem, stating the purpose and specific objectives, briefly describing the methodology, and providing an overview of the organization of the study.

Introduction

Quality manuals serve as the foundational documented information of a Quality Management System (QMS), providing the procedural and operational standards that guide office-level practices across an institution. In QMS-compliant organizations, these manuals are maintained under a controlled document framework, wherein only the latest approved version of a manual is made available for operational use, and any revision must pass through a defined review and approval process before it is formally adopted (ISO 9001:2015, Clause 7.5.3). In Leyte Normal University (LNU), the current practice for manual revision follows a controlled-copy workflow managed by the Quality Management System Office (QMSO ISO): when an office requires revisions to its manual, the QMSO ISO issues a controlled copy of that office's manual, the office performs its revisions independently, and the revised copy is returned to the QMSO ISO for review and approval before it is printed and integrated into the master copy repository. However, when this revision process is carried out through manual, paper-based handling, the review stage becomes difficult to execute consistently — since the QMSO ISO receives the revised copy without any structured change summary, reviewers must compare the old and revised versions entirely by hand to identify what sections were modified, a process that is time-consuming, prone to oversight, and difficult to document as part of a traceable approval record.

To address the challenges associated with manual document revision and version control in quality management environments, institutions have increasingly adopted electronic document management systems (EDMS) that centralize document lifecycle management — from creation and updating to approval, controlled release, and archiving. Gani et al. (2024) evaluated EDMS functionalities and found that core capabilities such as version control, automated workflow, security, and audit trails are significant contributors to overall information systems effectiveness in organizational settings. In a related case study, a link-based enterprise document management system was implemented to support ISO 9001 QMS compliance, wherein a versioning workflow — covering drafting, reviewing, and release — was established to ensure that each document update undergoes proper approval before becoming the current version, with minor and major changes handled through differentiated workflows. Revanasiddappa (2021) further explains that integrating AI-driven capabilities such as OCR into document management systems extends these systems beyond basic storage, enabling content-aware functionalities that support more effective document retrieval and handling. Despite these advancements, existing EDMS implementations primarily address document storage, retrieval, and general version tracking — the specific operational need of a QMSO ISO to systematically detect and present section-level differences between an old and a revised manual before approval remains largely unaddressed in many institutional implementations.

One key enabler for adding content-aware intelligence into a controlled document management system is the ability to extract usable, machine-readable text from documents. Optical Character Recognition (OCR) is a method for converting text in scanned pages or image-based documents into machine-readable text so it can be searched and processed (Laique et al., 2021). Technically, OCR engines analyze character and line patterns and output extracted

text — and when applied to quality manuals stored as scanned PDFs, OCR makes it possible to extract section content as plain text, which can then serve as the basis for automated change detection and full-text search within the system. On the other hand, Machine Learning (ML) can be defined as "a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty" (Broderick et al., 2023), which means ML systems improve their decisions by learning from data rather than relying on fixed, hand-crafted rules. Together, OCR and ML provide the foundational layer for a controlled document system to go beyond simple file storage — enabling the system to understand document content, suggest metadata tags for consistent indexing, and support reviewers in identifying meaningful changes between document versions.

Consistent with these definitions, several existing systems already apply OCR to convert document images into machine-readable text and then use ML models to categorize documents for more efficient handling in digital repositories and organizational records. In a study conducted by Luna et al. (2024), an automated document-classification pipeline for a university's repository integrated OCR and machine learning by distinguishing scanned from digital PDFs, applying OCR to scanned files, and classifying documents into predefined categories using models such as Random Forest and BERT. Similarly, Goodrum et al. (2020) applied OCR to extract text from scanned documents and evaluated machine-learning classifiers to categorize these files into document types, demonstrating how OCR-derived text features can support structured organization and downstream handling. In another related approach, Aydin (2021) performed OCR on document images and applied machine-learning techniques to classify documents based on the OCR-extracted text, further showing that digitized content can be transformed into usable signals for automated sorting and retrieval. Taken together, these studies

indicate that OCR combined with ML can reliably convert document content into structured information for automated classification and content-aware retrieval—capabilities that are essential foundations for intelligent indexing and review support in controlled document environments; however, many implementations remain centered on categorization and repository organization rather than providing a reviewer-ready, section-level comparison of revisions between an approved manual and a proposed update.

Despite the progress demonstrated by these OCR+ML systems, their scope remains largely centered on document classification and repository organization rather than supporting controlled-document review where approvers must validate revisions before an updated manual becomes the new master copy. In the study of Goodrum et al. (2020), the focus is on classifying scanned documents, and it also notes practical deployment challenges such as OCR noise and mixed-content files that can complicate consistent handling. Likewise, Luna et al. (2024) propose an OCR+ML pipeline to improve a university repository's categorization, but the work emphasizes classification performance and highlights issues such as class imbalance, document diversity, and OCR extraction quality as factors that can affect results, with broader deployment outside the study's scope. In a similar way, Aydin (2021) demonstrates OCR-based digitization and ML classification of document images, but the approach is presented mainly as categorization of digitized documents rather than a controlled-document approval process that must explicitly verify and justify revisions. Taken together, these limitations suggest that while OCR+ML is effective for extracting and structuring document content, there remains a gap in applying these capabilities to a controlled manual revision workflow—specifically, producing a reviewer-ready, section-level comparison between the current approved manual and a proposed update to support QMSO ISO approval decisions and master-copy updating.

However, current controlled-document revision practices in many institutions remain heavily manual, where proposed updates are reviewed by comparing printed or static copies without a structured, reviewer-ready presentation of what changed. This creates a practical gap for QMS offices: even when a revised manual is submitted, approvers still need an efficient way to identify modified sections, verify whether changes are compliant with established standards, and document the basis for approval before updating the official master copy. To address this gap, the study will develop a system that accepts a proposed revised manual, extracts machine-readable text through OCR when needed, and generates an automated, section-level comparison against the current approved version. The system will highlight additions, removals, and modified portions and produce a structured change summary to support faster, more consistent QMS ISO review, while maintaining traceability of revisions and approval actions as part of controlled-document management.

1.2 Purpose and Description

This study aims to develop a web-based controlled document management system, specifically for controlled manual revision and review support within Leyte Normal University that improves the efficiency, consistency, and traceability of QMS manual updates managed through the QMSO ISO. The system supports digital submission of proposed manual revisions through file upload and maintains a complete, time-stamped history of versions, review actions, and approval decisions to strengthen accountability during controlled-document updates. In addition, the system integrates OCR-based text extraction (for scanned or image-based pages) and automated comparison functions to generate a section-level change summary between the current approved manual and the proposed revision, enabling QMSO ISO reviewers to quickly identify what was modified prior to approval and master-copy updating.

To preserve document control and minimize erroneous or unverified changes, the system is designed as decision support: it does not automatically “approve” revisions, but instead produces reviewer-ready outputs (change highlights, affected sections, and comparison reports) that reviewers validate and confirm before a revision can be finalized and released as the new controlled version. Similar to the original structure where OCR and ML outputs are reviewed and confirmed by users before actions proceed, the proposed system emphasizes human confirmation for all approval and release steps.

The system will involve four primary user roles: (1) Manual Owner/Process Owner (office staff) who prepares and submits the proposed revision; (2) QMSO ISO Reviewer who evaluates the changes and records findings/comments; (3) Approver/Authorized Signatory who provides final approval for controlled release; and (4) System Administrator who manages user accounts, office mappings, access rights, and document control settings. The system will follow a structured workflow to ensure consistent revision handling: Manual Submission (upload revised file and encode basic details, select the current approved baseline), Text Extraction and Comparison (OCR when needed, section alignment, automated difference detection, and generation of a structured change summary), Review and Decision (QMSO ISO review, annotation/comment logging, revision return if required, and approval recording), and Controlled Release and Archiving (version incrementing, locking of approved copies, and audit-trail preservation for monitoring and reporting).

1.3 Objectives of the Study

Specifically, this study intends to:

1. Assess the performance of the OCR pipeline in extracting accurate machine-readable text from scanned/manual pages using standard OCR accuracy metrics such as Character Error Rate (CER) and Word Error Rate (WER).
2. Evaluate the accuracy of the automated section-level comparison and change-detection component (e.g., correctness of detected changed sections and highlighted edits) using appropriate agreement measures such as precision, recall, and F1-score against a reviewer-validated "ground truth."
3. Evaluate the developed system using black-box and white-box testing to verify that submission, comparison, review, approval, and controlled-release functions behave correctly across roles and scenarios.
4. Evaluate the developed system's quality using the ISO/IEC 25010 software quality model in terms of functional suitability, performance efficiency, usability, reliability, security, compatibility, maintainability, and portability.

1.4 Scope and Limitations

This study focuses on the development of a web-based controlled-manual revision and review support system for Leyte Normal University designed to handle QMS manual updates through paperless digital file intake (e.g., PDF upload) within a centralized platform. The system records revision lifecycle events using time-stamped logs to produce a traceable history of submissions, comparisons, reviews, approvals, and controlled release, and it generates section-level comparisons (e.g., highlighted changes and structured change summaries) between the current approved manual and a proposed revision to support QMSO ISO review and master-copy updating. The scope also includes role-based access, audit logging to strengthen

accountability, version history/archiving of approved copies, and basic monitoring/reporting views to improve visibility of revision status across offices and reviewers.

While the proposed system is designed to streamline controlled manual revision review and approval, it has several practical constraints. First, the accuracy of OCR-based text extraction and the reliability of comparison outputs depend on the quality of uploaded/scanned documents, so poor scans may reduce extraction accuracy and change-detection quality. Second, the system is assistive only and does not perform fully automated approval decisions; final approval and controlled release still require human confirmation to maintain accountability. Third, the system is designed for revisions processed within the platform and does not support capturing or tracking manual/offline revision exchanges; therefore, revisions handled outside the system may not appear in the audit trail and monitoring views. Fourth, the system relies on stable electricity and network connectivity for web-based operation, so interruptions may temporarily prevent submission, review, or approval actions. Fifth, monitoring and decision-support outputs depend on consistent user and office compliance in logging review comments, decisions, and acknowledgements; if these actions are not recorded, tracking views and generated metrics may be incomplete. Sixth, the study does not evaluate the legal or policy implications of document digitization and storage. Finally, if the system includes an internal electronic approval/acknowledgement feature, it functions only as an in-system workflow validation mechanism and is not a substitute for certificate-based digital signatures (PKI) or legal non-repudiation.

CHAPTER 2

Review of Related Literature

This chapter examines related literature and systems relevant to the study. It provides a contextual background from which the current study has been anchored on, with the goal of building upon existing knowledge and contributing new insights to the field.

Related Literature

EDMS for QMS, Approvals and Accountability

Electronic Document Management Systems (EDMS) are software solutions intended to capture, store, index, retrieve, and dispose of digital documents to streamline organizational workflows (Gamido et al., 2023). In higher education and similar organizational settings, EDMS are described as replacements for manual, paper-based handling that is prone to delays, loss, and physical storage constraints, using centralized repositories to manage institutional documents (Ayankoya et al., 2025; Pagayonan, 2022). In the Philippine SUC context, ISO 9001:2015 adoption is reported to be increasing to enhance organizational performance, yet maintaining QMS beyond initial certification can be challenging due to compliance deviations and lack of sustained resource support (Gamit et al., 2024). Taken together, these works suggest that EDMS is not merely a storage mechanism but a structural control environment that embeds procedural discipline into document movement, a feature particularly critical in sustaining ISO 9001-driven QMS practices (Gamit et al., 2024).

On approvals and accountability, the studies converge on workflow and routing as the backbone of EDMS-enabled governance. Ayankoya et al. (2025) illustrates workflow automation as a defined sequence of tasks where an initiator submits a document digitally, the system routes it to designated reviewers/approvers, returns it for modification when rejected, and forwards it to

the next authority when approved (Ayankoya et al., 2025). However, Yossy and Darmawan (2023) emphasize rule-based routing (predetermined approval providers and branching paths) to ensure documents follow a required chain and are less likely to be lost "in transit" (Yossy & Darmawan, 2023). Therefore, while their examples differ, both accounts frame accountability as something enforced structurally by embedding decision points and mandatory routes into the system itself.

These studies also argue about how deep accountability should go, contrasting role-based enforcement with highly granular traceability. Gamido et al. (2023) describes audit trails that record detailed access/viewer information, such as employee identity, timestamps, workstation name, and even MAC and IP addresses, and positions these logs as "proof of receipt" for distribution authenticity (Gamido et al., 2023). In contrast, Ayankoya et al. (2025) highlights security and responsibility enforcement through hierarchical roles (administrator, staff approvers, initiators), one-time tokens for staff registration, and e-signature integration to protect official actions (Ayankoya et al., 2025). Complementing these, version control is described as preserving document integrity by storing new versions during revisions while keeping the original accessible for reference, which supports reliable approvals over time.

Beyond routing, EDMS are also framed as improving operational efficiency through search/retrieval and records lifecycle governance (Gamido et al., 2023; Pagayonan, 2022). Gamido et al. (2023) and Pagayonan (2022) describe indexing-driven retrieval (tagging by reference number, author, subject, date range) and keyword-based querying to locate documents in real time rather than manual page-turning. Gamido et al. (2023) further extends EDMS value to disposal/retention management by assigning retention periods or disposal dates upon upload and using automated queries to identify records that have passed their "active until" date for

deletion or archival actions (Gamido et al., 2023). When aligned with evidence that SUCs face sustainability barriers in QMS implementation (resources, continuity, and compliance challenges), these EDMS capabilities collectively support the view that improved traceability, controlled access, and routable approvals can help stabilize QMS-related documentation work over the long term.

Despite these strengths, the EDMS studies largely emphasize document capture, storage, routing approvals, access control, audit trails, and general versioning, which improves accountability but still treats documents primarily at the "file and workflow" level. However, controlled QMS manual revision requires not only knowing who approved and when, but also enabling reviewers to efficiently validate what exactly changed between the current approved manual and a proposed revision in a reviewer-ready format. Therefore, this study is justified in developing a controlled document management system that strengthens QMSO review by producing structured, section-level change summaries (supported by OCR when documents are scanned), while preserving the traceability and accountability strengths already established in EDMS literature.

OCR in Document Management Systems

Optical Character Recognition (OCR) is consistently presented in the literature as the enabling layer that transforms scanned or image-based records into machine-readable text so that document systems can act on content rather than treating files as static images (Revanasiddappa, 2021). In Philippine university office contexts, within Leyte Normal University, Almenario et al. (2024) show OCR being used in a Records Management Office workflow to extract text from scanned administrative documents (Travel Orders, Office Orders, Special Orders, and circulars) and to auto-populate key database fields such as employee names, document subjects, and dates

of issuance (Almenario et al., 2024). However, Luna et al. (2024) highlight that at repository scale, OCR must be applied selectively, distinguishing digitally generated PDFs from scanned PDFs—because OCR is primarily necessary when text is otherwise inaccessible, making OCR an operational decision point in large classification pipelines (Luna et al., 2024). Therefore, across both local office workflows and large institutional repositories, OCR functions as a bridge that reduces manual encoding work by converting unstructured scans into usable text for downstream indexing and automation.

The studies also point to the technical reality that OCR quality depends heavily on document condition and preprocessing. Sancar et al. (2023) emphasize preprocessing (grayscale conversion, thresholding, noise reduction, and projection-based line/character separation) and even apply Levenshtein-based correction after extraction to mitigate recognition errors from scanning, while noting that high-DPI scanning devices correlate with better OCR performance (Sancar et al., 2023). Similarly, Luna et al. (2024) employ preprocessing through OpenCV (contrast adjustment, rotation correction, and DPI adjustment) before OCR, yet still report that complex layouts, noisy backgrounds, and multilingual content can lead to inconsistent extraction that affects downstream use (Luna et al., 2024). In contrast, Li et al. (2025) pushes the OCR pipeline toward layout-aware deep learning by incorporating a layout analysis stage (identifying titles, tables, and paragraphs) prior to recognition and producing editable outputs, positioning layout understanding as a solution to structure-related OCR limitations (Li et al., 2025). Taken together, these works suggest that OCR performance is not only a function of the recognition engine but also of scan quality and layout complexity, which means OCR-enabled document systems must anticipate extraction variability and include quality-aware preprocessing strategies.

At the same time, local implementations show that OCR does not eliminate the need for verification when source documents contain errors or constraints that OCR cannot resolve. Another system developed within Leyte Normal University was implemented by Balios et al. (2024) in an HRMO context handling orders, leave applications, and overtime requests, notes that OCR can produce analyzable and searchable text and can extract recipient names to support folder-based distribution, but it cannot detect or correct mistakes already present in the source document and physical-signature requirements limit the extent of full digitization (Balios et al., 2024). Almenario et al. (2024) similarly positions OCR as a response to time-consuming and error-prone manual entry, reporting through black-box testing that extraction can succeed even on complex formats and low-resolution documents, but still within the practical realities of office scanning workflows (Almenario et al., 2024). Therefore, OCR's contribution in office document management is best understood as decision support for data capture and indexing, powerful, but dependent on the quality and correctness of input documents and institutional constraints around document authenticity.

Despite demonstrating strong value for classification, and searchability, existing OCR-enabled systems primarily operate at the document-level rather than supporting structured, section-level comparison across controlled versions. For QMS manual revision, reviewers must validate precise additions, deletions, and modifications prior to approval. The literature reviewed does not explicitly address automated section-level semantic comparison between controlled versions, thereby justifying the proposed study's integration of OCR as a prerequisite for structured change summarization.

Machine Learning in Document Management Systems

Machine learning in these document pipelines generally appears as the component that converts OCR-extracted or digital text into structured signals for automated categorization, routing, and repository organization. According to Almenario et al. (2024), TF-IDF vectors derived from university records (Travel Orders, Office Orders, circulars) feed a Naive Bayes classifier that automatically assigns each document to a predefined records-management category, effectively reducing manual indexing work in a state university office. In contrast, Balios et al. (2024) process HRMO documents through tokenization, lemmatization, and TF-IDF before applying a Support Vector Machine (SVM) to classify them into HR-related classes such as Special Orders, Leave Applications, and Overtime Requests, reporting high user satisfaction with the accuracy and time savings in everyday HR workflows (Balios et al., 2024). Therefore, in these university office contexts, ML functions as a classification layer that sits directly after text feature extraction (with or without OCR) and drives automatic filing of documents into operationally meaningful categories.

Beyond office-scale systems, other studies extend ML usage toward routing, large-scale repository management, and layout-aware recognition. According to Sancar et al. (2023), multiple classifiers, including SVM, Naive Bayes, Random Forest, Logistic Regression, and an SGD classifier, are compared for petition routing, with SGD+BoW-TF-IDF achieving around 0.95 accuracy when deciding which administrative unit should receive a recognized petition, thus embedding ML as a decision point in workflow routing (Sancar et al., 2023). Similarly, Luna et al. (2024) evaluate Random Forest, SVM, and BERT-based models on 600,000 PDF documents, finding that a Random Forest over TF-IDF features outperforms transformer embeddings for classifying documents into 51 repository categories, which shows that, in some real-world institutional datasets, traditional ML with well-engineered features remains more effective than

deep language models (Luna et al., 2024). In a more technical direction, Li et al. (2025) employ YOLO-based layout analysis and CNNs to recognize structured printed documents, while Revanasiddappa (2021) discusses how deep learning and transformer models (CNNs, LSTMs, BERT, GPT) increase classification accuracy and reduce retrieval time in enterprise document management, albeit with high dependence on data quality and computational resources.

The preference for SVM and related classifiers in several of these systems is consistent with broader text-mining literature, which shows SVM to be particularly effective for high-dimensional text classification. In a text detection study, Reddy and Reddy (2021) report that SVM outperforms Naive Bayes, KNN, and other methods, achieving accuracies close to 97–98% in distinguishing spam, which they attribute to SVM's ability to find robust separating hyperplanes in sparse text spaces (Reddy & Reddy, 2021). Likewise, Soni (2019) finds that SVM delivers higher F-measure and accuracy than KNN on Scopus document data, especially when paired with appropriate feature extraction and techniques like bagging and boosting, further emphasizing SVM's strength in scenarios where TF-IDF features live in high-dimensional spaces (Soni, 2019). Taken together, these findings help explain why document-management–oriented systems such as Balios et al. (2024) select SVM for HR document classification: it offers strong accuracy, robustness in high-dimensional feature spaces, and good performance even on relatively modest datasets.

Despite these advances, the ML components across the reviewed studies largely focus on making document-level decisions, such as determining a document's type, its target office, or its repository category, rather than analyzing how the internal content of a controlled manual changes between revisions. Existing pipelines demonstrate that, once OCR has produced machine-readable text, ML can reliably classify and route documents, and even support semantic

search and layout-aware recognition, but they do not explicitly expose reviewer-ready, section-level change information needed in QMS manual approval workflows. In the context of QMS and ISO 9001:2015, where approvers must justify and document the basis of manual revisions before updating the master copy, this gap underscores the need for a system that repurposes OCR and ML not only for classification and routing, but for supporting fine-grained comparison of versions to assist human reviewers in controlled-document decision-making.

The reviewed literatures show that organizations, especially universities and SUCs, are increasingly combining EDMS infrastructure with OCR and ML to digitize records, automate classification and routing, and improve searchability and accountability, yet they still treat documents mostly as indivisible units in workflows. Together, these three bodies of work, such as EDMS infrastructure, OCR pipelines, and ML classification, provide the component foundations for the proposed system, while leaving the section-level comparison need unaddressed which is the QMS requirement to present reviewers with clear, section-level summaries of how a controlled manual has changed across revisions. Therefore, a controlled document management system occupies a distinct space in this landscape by leveraging the same OCR and ML foundations, yet directing them toward automated, reviewer-ready comparison of QMS manuals, rather than only toward document-level classification or routing. (International Organization for Standardization, 2015)

Related System

The systems presented in this section serve as foundational references for the proposed Controlled Document Management System for the Quality Management System Office of a Philippine state university. Each system illustrates how web-based platforms, and in some cases

OCR and machine learning, have been used to centralize institutional records, automate document handling tasks, and support ISO 9001:2015-aligned documentation practices in universities and large organizations.

Collectively, these related systems offer concrete design patterns and strategies for document storage, search, routing, and classification—ranging from general-purpose EDMS deployments in state universities to OCR- and ML-assisted petition and records-management solutions. By examining their architectures, workflows, and limitations, this study builds a clearer understanding of how existing systems handle document categorization and control at the file level, while motivating the need for a specialized controlled document management system that can support section-level comparison and reviewer-ready change summaries for QMS manuals.

Pagayonan (2022) developed an E-Document Management System for Northern Iloilo State University (NISU), a web-based repository aligned with ISO 9001:2015 that centralizes administrative, instructional, and personnel records under an N-tier architecture with presentation, business, data manipulation, and MySQL database layers, primarily serving the University Records Officer and staff as an official storehouse for institutional documents. It provides organized electronic storage and fast retrieval of diverse file types and manages the receiving, forwarding, and releasing of incoming and outgoing communications, thereby addressing earlier problems such as volume stacks of paper, missing master lists, and delayed forwarding of memoranda, although the documentation does not describe explicit document version control or section-level comparison features. Similar EDMS implementations in higher education report that centralizing records in a web-based system significantly improves document accessibility, reduces retrieval time, and supports quality management practices in

university operations (Bautista & Rosales, 2019). Evaluation using a developmental-descriptive design with 165 participants, a modified PSSUQ, and ISO/IEC 9126-based performance measures showed gains in organization, speed, accuracy, and reduced document loss compared to manual processes. However, the system functions as a broad administrative EDMS that focuses on whole-document recording and retrieval rather than generating reviewer-ready change summaries or enabling granular, section-by-section comparison of QMS manuals. Thus, while it demonstrates how a university can implement a web-based ISO-aligned EDMS, it does not yet meet the specific requirements of controlled QMS manual revision and detailed content auditing that the proposed system aims to address.

Almenario et al. designed a web-based Optimized Categorization of Records Management System for Leyte Normal University's Records Office, a web-based platform that automates document categorization, content extraction, and secure retrieval for administrators, secretaries, office heads, and university personnel through modules for dashboards, document scanning, advanced search, repositories, employee management, mail tracking, reporting, and system settings. It centralizes electronic documents in a database and uses OCR (PyTesseract) and a Naive Bayes classifier to auto-extract text and categorize records, supports quick and advanced search using rich metadata (document number, series, date, subject, employee), and enforces security through role-based access control, authentication, backup management, and audit trails of recent activities. This combination of OCR-based text extraction, TF–IDF vectorization, and machine learning models for document classification is consistent with broader research on automated text document classification pipelines (Almalki & Aljohani, 2025). The authors validated the system via white-box and black-box testing and a Likert-scale user evaluation with 165 respondents, reporting reduced human error, faster categorization and

retrieval, and improved accessibility by allowing remote access to records instead of in-person visits to the Records Office. Nonetheless, its automation focuses on single-document metadata extraction and classification rather than on tracking successive revisions of controlled manuals: it does not provide section-level content comparison, automated change summaries for reviewers, or an explicit lifecycle focus on QMS manuals and controlled copies. As such, it is closely related to the proposed system for its successful integration of OCR and ML in a Philippine university EDMS, but it remains a general administrative records system rather than a specialized controlled manual revision and review-support tool.

Sancar et al. proposed a web-based petition recognition and routing system for large-scale enterprises, tested using 1,421 petition samples from Ataturk University, to reduce delays, wrong-unit deliveries, and lost petitions by automatically directing electronic petitions to the appropriate administrative unit based on their textual content (Sancar et al., 2023). Petitions stored as images are scanned using Tesseract OCR, preprocessed (grayscale conversion, thresholding, noise reduction, line and character segmentation), corrected with the Levenshtein algorithm, and then vectorized via BoW–TF–IDF before being classified by different machine learning models (Sancar et al., 2023). Using 10-fold cross-validation, the best configuration—Stochastic Gradient Descent with BoW–TF–IDF—achieved 0.951 accuracy and 0.94 macro-F1 when routing petitions to eight destination units, and the authors highlight their work as one of the first to combine OCR, NLP, and ML for petition routing while also introducing a new Turkish petition dataset (Sancar et al., 2023). Despite its strong OCR-assisted routing capabilities and status tracking, the system operates on one-off short-form documents (petitions) and does not support versioned, section-level comparison of long-form controlled manuals or generate reviewer-ready change summaries in a QMS context.

Balios et al. developed a web-based Document Classification and Management System for the Human Resource Management Office (HRMO) of a state university in Tacloban City (implemented at Leyte Normal University) to address misplaced files, slow retrieval, and paper volume in HR processes (Balios et al., 2024). The monolithic system centralizes HR documents such as Special Orders, Office Orders, Travel Orders, various Leave Applications, and Overtime Requests, providing modules for digitized forms, files and folders, dashboards, analytics charts, and user management, with access controlled via roles (administrator, HR staff, faculty, MIS) and activity logs for monitoring usage (Balios et al., 2024). It offers modules for dashboards, profile management, folders/files, digitized leave and overtime forms, analytics charts, scanning, and user management, and it applies OCR together with an SVM-based ML pipeline (with tokenization, lemmatization, and TF-IDF) to classify documents while enforcing role-based access controls, password hashing, encryption, and audit trails via the Spatie RBAC library. Evaluated through the Waterfall SDLC, combined black-box and white-box testing, and ISO/IEC 25019:2023-based user surveys, the system was found to increase efficiency, speed up retrieval—with about 90% of respondents agreeing on faster access—improve accuracy in classification and leave/service credit computations, and mitigate misclassification and physical loss of paperwork. Still, its classification features operate at the document-type level and do not support section-level comparison, automated change summaries, or explicit tracking of controlled QMS manuals and their controlled copies; it also lacks content error detection, leaving quality control of text to manual checking. Thus, while it demonstrates a mature application of SVM and OCR in a university EDMS, it differs from the proposed system, which is explicitly oriented toward section-level revision analysis and controlled manual lifecycle management under a QMS framework.

Angala et al. developed a web-based Document Management System (DMS) for Ilocos Sur Polytechnic State College (ISPSC), Tagudin Campus, as a centralized digital environment to capture, store, organize, retrieve, and manage university files in response to the rapid growth of paper records and the resulting challenges in storage, searchability, and operational costs, targeting users such as the Campus Records Officer, faculty members, Deans, and College Secretaries, and handling documents that include memoranda from the President and VPAA, letter requests, transmittal letters, and prescribed ISO 9001:2015 QMS forms. The system is web-based with a mobile-first design, providing modules for secure login and role-based access control, user dashboards with analytics and reporting, document upload and tracking, search and filtering based on metadata like title, date, and source, report generation, and an activity log to monitor usage and support accountability. It functions as a centralized repository that improves retrieval speed and organization compared with the manual paper-based process, but the study does not describe an automated document version control mechanism or a formal, multi-stage digital approval workflow, and there is no integration of OCR or machine learning, with indexing relying on manually entered metadata (Angala et al., 2023). System evaluation using a descriptive-developmental design and the WAMMI questionnaire administered to 40 purposively selected respondents reported an overall satisfaction mean of 4.48 ("Very Satisfied"), with the highest scores for the system's ability to help users quickly locate documents. The use of standardized instruments such as WAMMI and PSSUQ to evaluate web-based information systems aligns with recent usability studies, which employ these questionnaires to quantify user satisfaction and perceived system usefulness (Pratama & Rachmawati, 2020). Consequently, while the ISPSC DMS offers an ISO 9001-aligned, campus-wide framework for managing documented information and reducing paper volume, it remains a general administrative archive

focused on whole-document storage and retrieval rather than a specialized controlled document management system capable of granular revision tracking, section-level content comparison, and reviewer-ready change summaries for QMS manuals.

All these related systems discussed above demonstrate how Philippine and international institutions have adopted web-based EDMS and OCR/ML-powered tools to centralize document storage, improve search and retrieval, automate categorization, and support routing and approvals, often in the context of ISO 9001:2015-aligned documentation. Nonetheless, they primarily treat documents as whole units for storage, filing, or routing, without addressing the specific controlled-manual requirement of presenting QMS reviewers with automated, section-by-section change summaries between the current approved manual and a proposed revised version, a gap that the proposed Controlled Document Management System for the QMS Office aims to address.

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